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Dr. Dilshan Benaragama,
Digital Agronomy and Weeds Lab (DAWL),
University of Manitoba, Winnipeg, Canada,

Dear Dr. Benaragama,

I am writing to express my strong interest in M.Sc. opportunities in the Digital Agronomy and Weeds Lab at the University of Manitoba. I was unable to submit my application directly through the "Apply Now" button on the lab website, so I am reaching out to you by email instead. I am Henry Mashen Masanja, a final-year student in Agricultural Sciences and Engineering at EARTH University in Costa Rica, graduating with a cumulative GPA of 3.6/4.0. Your lab's integration of UAV-based sensing, remote sensing, and data-driven decision support to bridge traditional weed science with digital agriculture aligns closely with my research background and interests.

My BSc thesis, "Estimating Aboveground Biomass of Kernza Using UAV Multispectral Imagery and Machine Learning," demonstrates direct experience with the core methods central to DAWL's research. I design and execute UAV data collection campaigns, preprocess high-resolution multispectral and RGB imagery, perform image segmentation and feature extraction, and develop machine learning models, including random forest, support vector machine, and XGBoost, validated against ground-truth field measurements. This workflow closely parallels the UAV-based crop monitoring research being conducted by Pantha in your lab, and the underlying machine learning and image-analysis skills transfer directly to the kind of computer vision and data analytics approaches used in machine learning for weed detection, such as Shirmith Nirmal's work.

I am proficient in Python for machine learning and geospatial automation, experienced with QGIS, Pix4D, and Agisoft Metashape, and comfortable handling large UAV image datasets and building reproducible processing pipelines. My 2025 research internship at Aarhus University in Denmark strengthened my experience with field data collection and interdisciplinary collaboration. My background in Agricultural Sciences and Engineering provides a solid foundation in crop production and field-scale variability, complementing DAWL's focus on precision agriculture and digital weed management.

I am particularly drawn to DAWL's mission of bridging scientific innovation with practical, farmer-facing applications, and to the lab's combination of UAV platforms, remote sensing, and machine learning for site-specific, sustainable crop management. I would welcome the opportunity to contribute to ongoing research in UAV-based monitoring or machine learning applications for weed and crop management, while developing deeper expertise in digital agronomy under your supervision.

Thank you for considering my application. I would welcome the opportunity to discuss how my background can contribute to the Digital Agronomy and Weeds Lab. I look forward to hearing from you.

Best regards,

Henry Mashen Masanja